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RESEARCH ARTICLE

A Subject-Independent Machine-Learning Framework Using Multimodal EEG Features for Identifying Learning Disorders in Children

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ABSTRACT Developmental learning impairments substantially hinder academic progress and result in psychological and social challenges. They often remain undiagnosed due to the unreliability and scalability of diagnostic tools. Non-invasive neuroimaging tools, such as EEG, provide insight into the neural mechanisms underlying various disorders. This study introduces an AI-driven, subject-independent, and multimodal EEG-based diagnostic framework for the objective classification of learning disorders in children, contributing to the advancement of intelligent neurotechnological tools for cognitive healthcare. EEG was conducted on 146 children during specifically designed non-numerical and numerical working memory tasks. Features were extracted from ERPs, spectral dynamics, cross-frequency coupling, and behavioral data. Dimensionality reduction was performed using three complementary feature selection strategies, and classification was implemented with a Bayesian-optimized SVM. The highest classification accuracy, reaching 86.36%, was achieved in the negative condition of the numerical task using a combination of ERP and non-ERP features. ERP analysis revealed that the LPN component differentiates between groups. Behavioral results indicated that the control group responded more quickly and was more accurate in both numerical and non-numerical tasks. Integrating behavioral variables with EEG features did not yield additional gains, indicating that neural markers provided the dominant discriminative power. This work demonstrates the feasibility of AI-based, neurotechnology-enhanced intelligent diagnostic systems for learning disorders. The integration of multimodal EEG features into a scalable, reproducible, and subject-independent pipeline provides a promising step toward the practical clinical translation of automated screening tools for developmental cognitive disorders.

INDEX TERMS EEG, event-related potentials, intelligent diagnostic systems, learning disorders, machine learning, multimodal features.

I. INTRODUCTION

Childhood cognitive development is essential for academic success, emotional well-being, and social integration. Neurodevelopmental disorders such as dyslexia and

dyscalculia can disrupt this developmental trajectory, leading to educational and psychological problems. Dyslexia, which affects approximately 7.1% of primary school-aged children, is characterized by challenges in letter recognition, decoding, and word recognition. Dyscalculia, which is estimated to affect 3–6% of the population, is characterized by impairments in numerical cognition, including arithmetic

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processing and number sense. These disorders can persist into adolescence and adulthood, causing academic underachievement, low self-esteem, and school dropout [1], [2].

Neuroimaging studies have revealed the distinct neural mechanisms underlying the disorders. Dyslexia disturbs white matter pathways, particularly the superior longitudinal fasciculus and corpus callosum, which are needed for reading and phonological processing [3]. In contrast, dyscalculia is linked to stronger connections in the intraparietal sulcus and fusiform gyrus when doing math, indicating that the problem lies in how numbers are understood rather than in the use of these areas less [4]. Due to a lack of awareness and screening methods, many children with these diseases go untreated or receive delayed treatment. Neurophysiological approaches, such as Electroencephalography (EEG), are being used to address these issues. EEG is a safe, non-invasive approach that has an excellent temporal resolution and is widely available. Event-Related Potentials (ERPs) are derived from EEG signals. Researchers commonly use ERPs to study memory and attention, as they reflect the various stages of neural processing in the brain [5].

Turri et al., focusing on dyslexia, demonstrate that dyslexia's altered aperiodic EEG activity and diminished beta rhythms [6]. Formoso et al. reveal that dyslexic children differ from typically developing peers in phase synchrony and network efficiency, especially during auditory tasks [7]. In addition to Working Memory (WM) and phonological processing disruptions, dyscalculia also affects spatial number representation and numerical reasoning [8]. ERP analysis revealed a decrease in N400 and Late Positive Component (LPC) in cases of dyscalculia. These findings indicate arithmetic reevaluation difficulties. On the other hand, dyslexia has disturbed N320 and LPC responses connected to phonological learning [9], [10]. Different machine learning approaches have employed EEG/ERP processing for dyslexia classification, achieving 96% accuracy [11], [12]. Combining EEG with other variables, such as eye-tracking and demographic data, made the diagnosis more accurate [13]. Seshadri et al. investigated 30 participants (15 dyslexic, 15 controls). They employed EEG and graph theory to study brain networks during arithmetic exercises. Delta band connectivity increased, especially in T3, O1, and F8. The study noted reduced connectivity in the theta, alpha, beta, and gamma bands across several regions, including P3, Pz, T5, and prefrontal areas [14]. In another study [15], the same research team proposed an EEG-based framework for detecting Learning Disabilities (LD) in children aged 8–16, including 20 LD and 20 non-LD participants. Digital Wavelet Transform (DWT) was used to calculate EEG features, and a filter-based feature selection method was applied. The maximum classification accuracy reached 97.5%. Oliaee et al. [16] employed EEG to study dyslexic children receiving transcranial Direct Current Stimulation (tDCS) and found brain activity differences before and after treatment. Principal Component Analysis (PCA) and

Sequential Floating Forward Selection (SFFS) were used to choose important features related to frequency and phase, achieving 92% accuracy in classifying treatment responses with a Support Vector Machine (SVM). Kumar et al. [17] employed DWT to extract EEG features from 8–16 year olds. They found that left temporal electrodes detect learning disorders with 94.4% accuracy using RBF-SVM. Mikheev et al. in a 2024 preliminary study [18] used EEG data from 65 children to assess four machine learning algorithms for dyslexia classification: SVM, Ensemble, Discriminant Analysis, and K-Nearest Neighbor (KNN). KNN had the highest testing accuracy (69.5%) with 5-fold cross-validation on a 40/20 train-test split, followed by SVM (59.9%). A confusion matrix study showed KNN and SVM were best for dyslexia detection. A recent study [19] examined dyslexic children's memory problems in 34 pupils (17 dyslexic, 17 control). The Opposite Meanings task measured long-term memory, and the Backward Digit Span task assessed WM. Dyslexic children performed worse in both memory domains than controls. Nakamura et al. employed EEG to examine how WM restrictions affect syntactic and semantic processing during a sentence judgment task. ERP results demonstrated decreased syntactic sensitivity under WM stress, as P300 responded less to syntactic errors. WM limitations did not alter N400 responses to semantic errors, suggesting that semantic processing is more resilient [20]. Martínez-Briones et al. studied resting-state EEG rhythms and IQ in children with Learning Disorders (LD). EEG and WISC-IV data from 91 LD and 45 (Typical Development) TD children were evaluated. High delta and theta power were negatively correlated with IQ, while frontoparietal gamma activity was positively correlated. When combined with IQ scores, EEG features explained 13.1% of IQ variance and distinguished LD from TD children with 82.4% accuracy [21]. A recent study [22], Manimaran et al. proposed a hybrid Random Forest-XGBoost dyslexia classification model. The dataset had numerical and categorical characteristics treated with one-hot encoding, oversampling the Minority Class (SMOTE) for class imbalance, and PCA for dimensionality reduction. Random Forest initially outperformed other models with 95.31% accuracy, but the hybrid Voting Classifier increased to 96.23% accuracy with F1-scores and precision reaching 0.96, exhibiting good and consistent dyslexia identification. Despite the increasing research on EEG-based diagnostic methods, methodological and conceptual limitations persist. The field's inconsistency in experimental design, data preprocessing, and feature selection hinders reproducibility and cross-study comparisons [23]. Additionally, most EEG and ERP studies have focused primarily on dyslexia, leaving dyscalculia—and especially the comorbid occurrence of both disorders—largely underexplored [24], [25]. Furthermore, although multimodal frameworks incorporating auxiliary measures such as handwriting kinematics and eye-tracking have been proposed to enhance classification performance in dyslexia, these methods have not yet been systematically applied to dyscalculia [26], [27].

The present study addresses these gaps by introducing a data-rich and multidimensional framework specifically designed for the objective assessment and classification of Specific Learning Disorders (SLD). A Working Memory (WM) task was developed, incorporating both visuo-spatial and verbal components, and systematically varying cognitive load through numerical and phonological stimuli. By focusing on children aged 6 to 11, the study provides insights relevant to early developmental stages, extending beyond the typical age ranges examined in prior research. A key innovation of this work is the simultaneous collection of behavioral and neural (EEG) data, time-locked to the same cognitive process, creating a multilevel feature space. By integrating cognitive neuroscience, developmental psychology, and machine learning. The methodology is low-cost, scalable, and compatible with real-world clinical settings, positioning it as a significant step toward early, objective, and comprehensive neurocognitive screening for learning disorders.

The main contributions of this study can be summarized as follows: (1) a multimodal framework integrating ERP, full-spectrum EEG, and behavioral features within a unified pipeline, providing a more comprehensive representation compared to existing single-modality EEG-based approaches; (2) a subject-independent and leakage-free evaluation strategy to ensure realistic and generalizable performance, addressing the limitations of subject-dependent evaluation commonly reported in prior studies; (3) a systematic comparison of multiple feature selection techniques under a consistent experimental setup, which is often lacking in existing work where methods are evaluated in isolation; and (4) the design of a working memory paradigm incorporating both numerical and non-numerical conditions to capture complementary cognitive processes associated with different learning disorders. These contributions distinguish the proposed approach from existing EEG-based classification studies, which are often limited to single-modality features or subject-dependent evaluation settings.

II. DATA RECORDING

A. PARTICIPANTS

This case-control study included 146 children aged 6–11 years (mean age = 8.69 ± 1.40 ; 86 males, 60 females), comprising 53 with dyslexia and/or dyscalculia and 92 typically developing controls. Groups were matched by age, gender, and school to minimize potential confounding factors. All children were tested in urban centers (Karaj and Tabriz, Iran). Families represented the typical urban population, with parental education generally ranging from secondary school to university level. Participants were predominantly of Persian and Azeri ethnolinguistic backgrounds, reflecting the demographics of the recruitment regions. Diagnostic assessment combined standardized tools: the Child Symptom Inventory-4 (CSI-4) [28], a mathematics test tailored for dyscalculia—the Iranian KeyMath adaptation [29], the Colorado Learning Disabilities Questionnaire

(CLDQ) [30], teacher questionnaires on reading and writing, the Wechsler Intelligence Scale for Children-IV (WISC-IV) [31], and a structured clinical interview based on DSM-5 criteria. Exclusion criteria included a Full-Scale IQ below 80, neurological or psychiatric disorders, history of brain injury, or use of medications affecting cognition. All procedures followed the Declaration of Helsinki and were approved by the National Institute for Medical Research Development (IR.NIMAD.REC.1402.013). Written informed consent was obtained from parents or guardians.

B. EXPERIMENTAL DESIGN

A Working Memory (WM) paradigm was developed using PsyTask software (Mitsar Ltd.), consisting of two carefully matched blocks eliciting non-numerical and numerical WM processes. Each trial began with a 200 ms fixation cross, followed by a 3,000 ms encoding phase in which a 2×2 matrix of two stimuli per cell was presented. After a 500 ms visual mask, the corresponding spoken labels for the items in the attended location were played, allowing the child to compare the auditory information with the visual item from the encoding matrix. In addition, each trial included an auditory presentation of stimulus names lasting approximately 2,000 ms, followed by a 1,200 ms auditory mask. In the 3,500 ms decoding phase, participants judged whether the encoded and decoded matrices were identical in both content and spatial arrangement (right button = match, left button = mismatch).

In the non-numerical block (Figure 1), participants viewed pairs of familiar one-syllable object pictures and heard their corresponding spoken names. In the numerical block (Figure 2), they were presented with pairs of dot arrays (1–4) and their respective number names. In both tasks, trials were classified as positive when encoding and decoding arrays matched in both aspects, and negative otherwise. Each task comprised 70 trials, including six practice trials, and task order was counterbalanced across participants. Behavioral data included both accuracy and mean reaction time, with discrimination ability quantified using d' ($z[\text{Hit Rate}] - z[\text{False Alarm Rate}]$).

The dataset includes 3-minute resting-state EEG recordings under both eyes-open and eyes-closed conditions collected before the WM tasks. EEG-ERP recordings were obtained in a quiet, shielded environment using a 19-channel elastic Electro-Cap (10–20 system, Ag/AgCl electrodes; referenced to the earlobes, ground at AFz). Impedance was maintained below 5–10 k Ω with conductive gel. Signals were sampled at 500 Hz via a DC Mitsar™ amplifier. The WM experiment consisted of two ~15-minute sections (numerical and non-numerical), randomized to avoid order effects, with short breaks provided between blocks. The total recording session lasted approximately one hour. The experimental paradigm, data acquisition protocol, and the dataset used in this study are the same dataset previously published in [32] which was collected by our research team. The present work

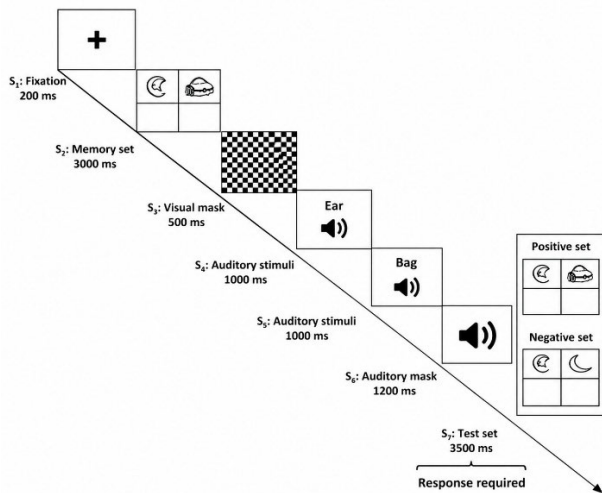


FIGURE 1. Experiment of non-numerical WM task [32].

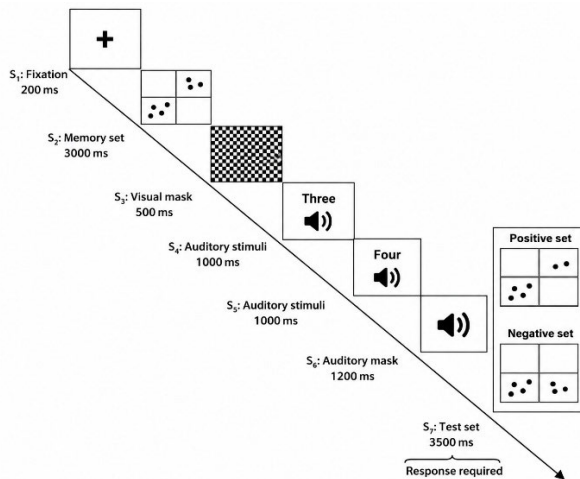


FIGURE 2. Experiment of numerical WM task [32].

builds upon this dataset by introducing a new analysis and classification framework.

III. MACHINE LEARNING PIPELINE

An overview of the proposed EEG-based multimodal classification framework is illustrated in Figure 3.

A. EEG PREPROCESSING

The preprocessing of EEG data was conducted using MATLAB and the EEGLAB toolbox, adhering as closely as possible to the MAKATO protocol [33]. First, the raw EEG signals were bandpass filtered using zero-phase Finite Impulse Response (FIR) filters with a bandwidth of 0.5-40 Hz. This step removed low-frequency drifts and high-frequency muscle and electrical noise, including power line interference and harmonics. Artifact Subspace Reconstruction (ASR) was employed to eliminate high-amplitude, transient noise. The ASR algorithm decomposes short sliding

windows of EEG data using PCA, comparing them to a clean calibration signal and reconstructing segments that significantly deviate [34]. The ASR burst detection threshold was tested in the range of 10–25 and set to 15, balancing artifact rejection and signal preservation.

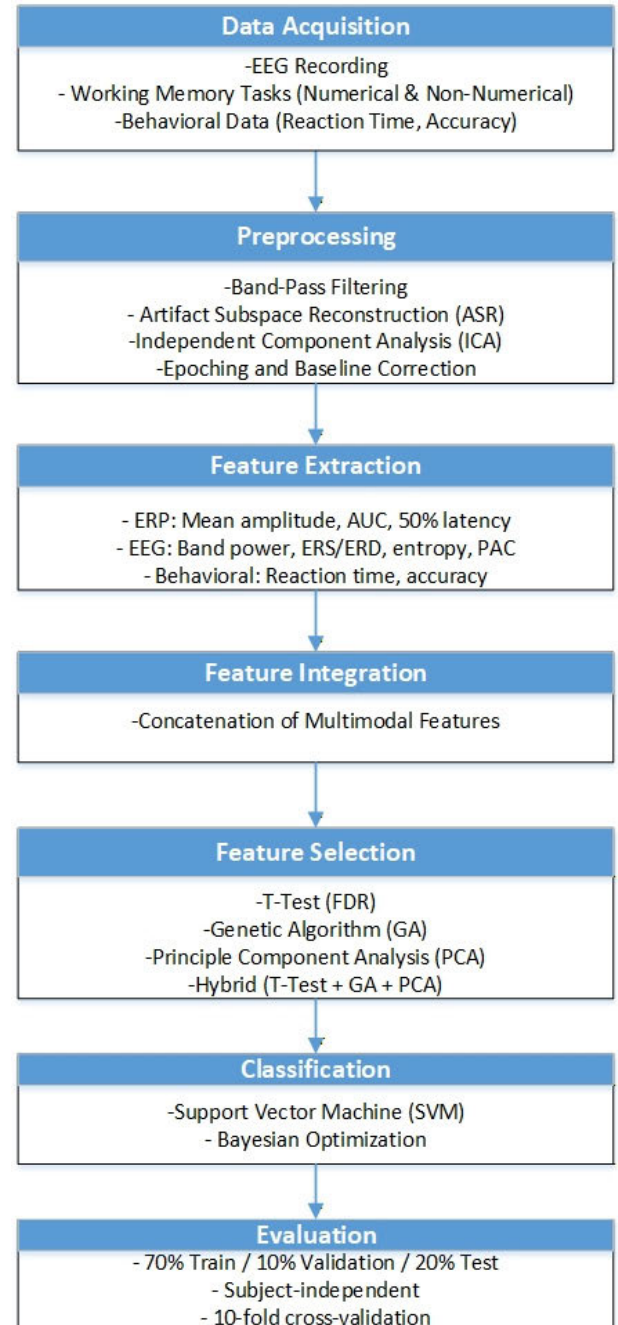


FIGURE 3. Overview of the proposed EEG-based multimodal framework.

To further refine signal quality, Independent Component Analysis (ICA) was performed. This mix source separation (MSS) method helps isolate and remove non-neural components such as ocular, muscular, cardiac, and line

noise artifacts. Components were classified using the ICLabel plugin [35], a neural-network-based tool trained on over 200,000 components. Only components with a brain probability >60% were retained for reconstruction; others were excluded. To ensure objectivity and reproducibility, this procedure was implemented consistently and uniformly across all subjects through an automated pipeline, minimizing human intervention and bias. Finally, after preprocessing, there were 92 participants (54 male) in the control group and 54 participants (32 male) in the disorder group, whose diagnoses included dyscalculia, dyslexia, or both. After ICA, ERP extraction focused on working memory-related components, with epochs time-locked to stimulus onset and spanning from 100 ms pre-stimulus to 1000 ms post-stimulus. Final epochs were baseline-corrected and visually inspected to ensure sufficient signal quality. The preprocessing pipeline followed a structured sequence, including bad-channel rejection, channel interpolation using spherical splines, re-referencing to the common average, artifact correction using ASR, ICA-based component cleaning, and final epoching and rejection [33].

B. FEATURE EXTRACTION

After preparing the EEG data, a clear process was set up to gather a wide variety of features from (1) ERP signals, (2) behavioral data, and (3) full-band EEG characteristics. This comprehensive feature engineering strategy aimed to capture localized and distributed neural dynamics relevant to the diagnosis of learning disorders.

ERP-Based Features: The initial stage focused on ERP features extracted from statistically significant time windows identified via False Discovery Rate corrected (FDR) t-tests across EEG channels. For each component of interest, three primary features were calculated per channel: (1) mean amplitude, reflecting the average voltage across the component window; (2) latency to 50% of the Area Under the Curve (AUC), providing a timing index of peak neural engagement; and (3) total AUC, capturing the overall strength of the response.

Behavioral Features: Two behavioral features—reaction time and response accuracy—were computed and normalized across trials to enrich the neural data with task-level cognitive performance. These indices serve as real-time proxies of cognitive load and task engagement, allowing us to examine their alignment with neural patterns.

EEG Features: In addition to ERP-derived markers, a broader feature extraction step across the entire ERP time window was implemented, designed to quantify both spectral and dynamical signal characteristics: (1) Time-domain: AUC to capture signal magnitude. (2) Frequency-domain: Band power estimates for delta, theta, alpha, beta, and gamma bands, along with power ratios such as theta/alpha and alpha/beta, known to reflect attentional and cognitive control processes. The frequency bands were defined as delta (0.5–4 Hz), theta (4–8 Hz), alpha (8–13 Hz), beta (13–30 Hz), and gamma (30–40 Hz). Spectral power was computed

via Welch's method using Hamming windows with 50% overlap, producing robust power estimates for each channel and condition.

Event-Related Synchronization/Desynchronization:

The ERS/ERD metrics are used to detect transient power modulations associated with stimulus processing. The formula is as follows:

$$100 * \frac{P_{active} - P_{baseline}}{P_{baseline}} \quad (1)$$

Which P_{active} and $P_{baseline}$ corresponds to the power of the active and baseline periods, respectively [36]. The active period (8.4–10.0 s) corresponded to neural processing following the retrieval probe presented at 8.3 s, covering the working-memory retrieval and decision interval. The baseline period (7.4–8.4 s) captured pre-probe activity.

Entropy: Shannon entropy was used to characterize the signal complexity and neural variability, often linked to cognitive flexibility.

Phase-Amplitude Coupling (PAC): To capture cross-frequency dynamics, PAC between theta-phase and gamma-amplitude was extracted. This coupling reflects cognitive coordination and has been linked to processes such as memory and attention. PAC was calculated using the Modulation Index (MI):

$$MI = \left| \frac{1}{N} \sum_{n=1}^N A(n) e^{i\Phi(n)} \right| \quad (2)$$

where $A(n)$ is the amplitude of the gamma band, and $\Phi(n)$ is the phase of the theta band at time n . The term N refers to the number of time points in the signal. EEG signals were bandpass filtered, and the Hilbert transform was applied to extract amplitude and phase [37]. PAC was computed per channel over the ERP window, producing one feature per channel.

C. FEATURE SELECTION

A multi-tiered feature selection strategy was implemented to mitigate overfitting, reduce redundancy, and improve classifier generalizability in the context of high-dimensional EEG and behavioral data. This was crucial given the combination of ERP-derived features, full-spectrum signal metrics, and behavioral features. Each method was applied within a rigorously controlled evaluation pipeline involving 10 repeated random subject-independent splits, each using a different random seed (positive, negative, combined) and Bayesian-optimized SVM classifiers [38]. Feature selection was applied solely to the training set in each repetition to prevent data leakage.

(1) T-test: A univariate statistical method was used to rank features based on between-group differences, applying FDR correction to control for multiple comparisons.

(2) Genetic Algorithm (GA): A wrapper-based optimization technique that explores feature subsets through biologically inspired operations. GA accounts for feature

interdependencies and is well-suited for non-linear problems. For the genetic algorithm, a population size of 50 and 50 generations were used with a crossover rate of 0.8 [39].

(3) Principal Component Analysis (PCA): An unsupervised dimensionality reduction method used to transform original features into a smaller set of orthogonal components. Components were retained to preserve $\geq 95\%$ of the total variance [40].

(4) Hybrid Approaches: To leverage the strengths of both filter and wrapper methods, two-stage combinations were also explored:

GA + PCA: GA-selected features were further transformed via PCA to enhance variance representation while maintaining optimized subset selection.

T-test + GA + PCA: A comprehensive pipeline where initial statistical filtering was followed by GA-based refinement and final PCA transformation to reduce redundancy and ensure compact, informative feature spaces. This sequence was selected to first mitigate overfitting risk (T-test), then capture nonlinear feature interactions (GA), and finally stabilize the feature space and enhance generalization by removing redundancy (PCA).

Each feature selection method was exclusively applied to training data in a cross-validation-safe manner to prevent data and validation leakage. Their impact on model performance was assessed over 10 randomized repetitions (K-fold cross-validation), capturing variability and robustness.

D. CLASSIFICATION PROCEDURE

The classification stage used SVM with an RBF kernel, which was selected because it can handle complex relationships and large sets of features. To optimize model performance, kernel width and regularization parameters were fine-tuned through Bayesian optimization, ensuring convergence on hyperparameter settings that reduce generalization error [38]. To thoroughly assess classifier performance and prevent overfitting, the dataset was split into 70% for training, 10% for validation, and 20% for testing. A 10-fold cross-validation method was applied during training to evaluate within-sample variability and enhance robustness. Importantly, all preprocessing, feature selection, and hyperparameter tuning were performed only on the training folds, with no access to the validation or test sets, to ensure a fully leakage-free and subject-independent evaluation. Due to the variety of feature vectors generated in earlier stages, classification scenarios varied based on the input configuration. These included: (1) ERP-derived features from a single component, (2) combined ERP features from two components (early perceptual and late cognitive), (3) ERP combined with behavioral features, (4) full EEG signal features along with ERP and behavioral features, and (5) features reduced using the mentioned methods and selected through hybrid pipelines. Each scenario allowed for assessing how different neural and cognitive signals contribute to classification performance under

varying task conditions (positive vs. negative; numerical vs. non-numerical).

IV. RESULTS

The study aimed to distinguish children with LD (dyslexia or dyscalculia) from typically developing peers using EEG/ERP features and behavioral performance during numerical and non-numerical WM tasks. This section presents a detailed account of the experimental findings derived from demographic analyses and EEG-based classification pipelines.

A. DEMOGRAPHIC AND ERP COMPONENT ANALYSIS

A comprehensive analysis was conducted on both the control group and the children with LD to eliminate any potential influences of age and gender. In particular, ERP components were examined to assess their sensitivity to stimulus type, age brackets, and gender differences.

Control Group: The study investigated ERP responses in the control group under two stimulus categories (numerical and non-numerical) and two different conditions (positive and negative). Components like P100 and P300 did not alter much between conditions. Non-numerical-based trials altered Late Positive Negativity (LPN), particularly at the O1 electrode (Figure 4a). This 500–800 ms post-stimulation phase is often related to emotion processing and motivation. It is important to note that in this paradigm, the retrieval probe appears at 8300 ms within each trial; therefore, all subsequent ERP analyses and component windows (e.g., 8.4–8.7 s and 8.9–9.4 s) are time-locked to the probe rather than the initial encoding stimulus. Note that the positive LPN amplitude was more negative than the negative one. This reversal suggests that identifying known visual stimuli (positive set) may improve evaluation or motivation due to their prominence or cognitive accessibility.

To examine age effects, participants were divided into three developmental cohorts: 6–7, 8–9, and 10–11. After conducting an ANOVA followed by the Bonferroni correction to statistically evaluate these subgroups, the observations confirmed that there were no significant differences in the form or magnitude of the ERP among the conditions. Gender analysis showed no significant differences.

Disorder Group: The same methodology was implemented to analyze ERP data from the LD group. Particularly during verbal activities, the LPN component demonstrated a substantial disparity between positive and negative conditions. This finding suggests that this ERP component may be susceptible to changes in the cognitive processes of individuals with LD (Figure 4b). As in the control group, no significant ERP differences were observed between genders or age groups. However, diagnostic status was the primary factor contributing to ERP variation in this sample, rather than demographic variables.

B. STATISTICAL INSIGHTS AND BEHAVIORAL DATA

To identify time intervals that exhibited substantial disparities between the two groups, paired t-tests with FDR correction

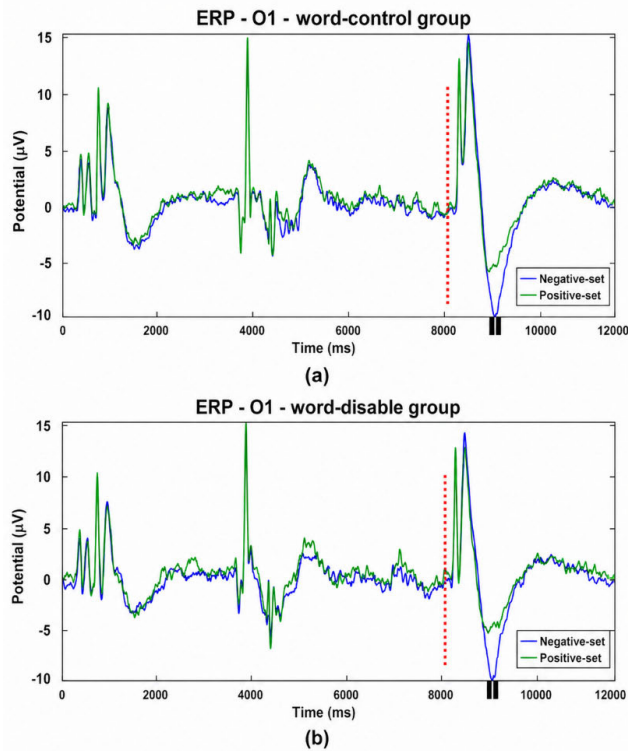


FIGURE 4. ERP signal at channel O1 (a: control group, b: disorder group) during the non-numerical task. The green plot corresponds to the positive set, and the blue one displays the negative set. The stimulus appears at 8300 ms (indicated by the vertical red line). A Bonferroni-corrected t-test ($p < 0.05$) reveals a significant difference at approximately 9300 ms, corresponding to the LPN component.

were conducted on ERP data from all EEG channels before classification. The O1 channel in Figure 5 illustrates that the control and disorder groups were consistently distinguished by varying time intervals, particularly during the LPN period. Consequently, these intervals were considered to identify ERP components for feature extraction. Furthermore, the behavioral data that was gathered during the assignment was analyzed. Two features, (1) response time, which assigns higher scores to responses that are completed more quickly, and (2) normalized correctness, which quantifies the discrepancy between correct and incorrect responses, were reviewed. The control group consistently outperformed the experimental group across both parameters, irrespective of the stimulus type or environment, as indicated by the box-plot graphs (Figure 6). However, the limited added value of behavioral features in classification suggests that neural markers captured by EEG provided more direct and sensitive indicators of group differences than overt task performance. It is worthwhile mentioning that although the O1 channel is presented here as a representative example, all EEG channels were included in the feature extraction and classification process, and no single channel was treated as the primary channel.

C. MACHINE LEARNING CLASSIFICATION

Multiple machine learning pipelines were conducted to classify individuals into control and disorder categories. Each participant was provided with ERP data from 19 EEG channels for both positive and negative conditions, together with a concatenated status, for two tasks: one focused on numerals and the other on words (Figure 7). It is important to note that, to prevent validation leakage and ensure subject-independent classification, the dataset was randomly partitioned into three distinct subsets: training, validation, and test sets, as described previously. The test set remained completely unseen throughout all stages of feature selection, model optimization, and training, and was only utilized for final performance evaluation. To enhance the robustness of the results and mitigate variability due to random partitioning, the entire classification procedure was repeated across 10 independent iterations.

(1) Single ERP Component: The initial classification model utilized features from a single ERP component, which were selected following the statistical study mentioned earlier. Accordingly, the first ERP analysis window, ranging from 8.4 to 8.7 seconds post-stimulus, was selected to capture early cognitive responses related to stimulus encoding, attentional allocation, and initial working memory engagement. The mean amplitude, 50%, and total AUC of each channel within the component's time window were calculated. The non-numerical task exhibited the highest accuracy when all conditions were combined, with classification accuracies ranging from 60.91% to 70.00%. It is crucial to emphasize that specificity was consistently greater than sensitivity, indicating that the classifier was more effective in identifying healthy children than those who had LD. The classification results are illustrated in Table 1.

Two additional behavioral features were incorporated into the ERP-derived vectors to determine whether behavioral data could assist in classification. The results indicated that the accuracy of classification improved slightly in the majority of settings. The accuracy of the numerical task increased to $77.27\% \pm 5.75\%$ in the positive condition. However, the sensitivity remained low, indicating that the added behavioral factors did not significantly assist the classifier in identifying impacted children. As illustrated in Table 2, the results are presented.

(2) Two ERP Components: A second ERP component was incorporated due to its importance, as a subsequent time frame likewise demonstrated significant disparities across the groups. The second window, spanning 8.9 to 9.4 seconds, was designed to assess late-stage neural activity associated with retrieval, decision-making, and cognitive reallocation. The three identical features were computed. The numerical task showed substantial enhancement by the implementation of this dual-component feature set for classification, especially in the negative condition, achieving an accuracy of $80.91\% \pm 7.82\%$. Furthermore, the sensitivity rose to 0.70, signifying

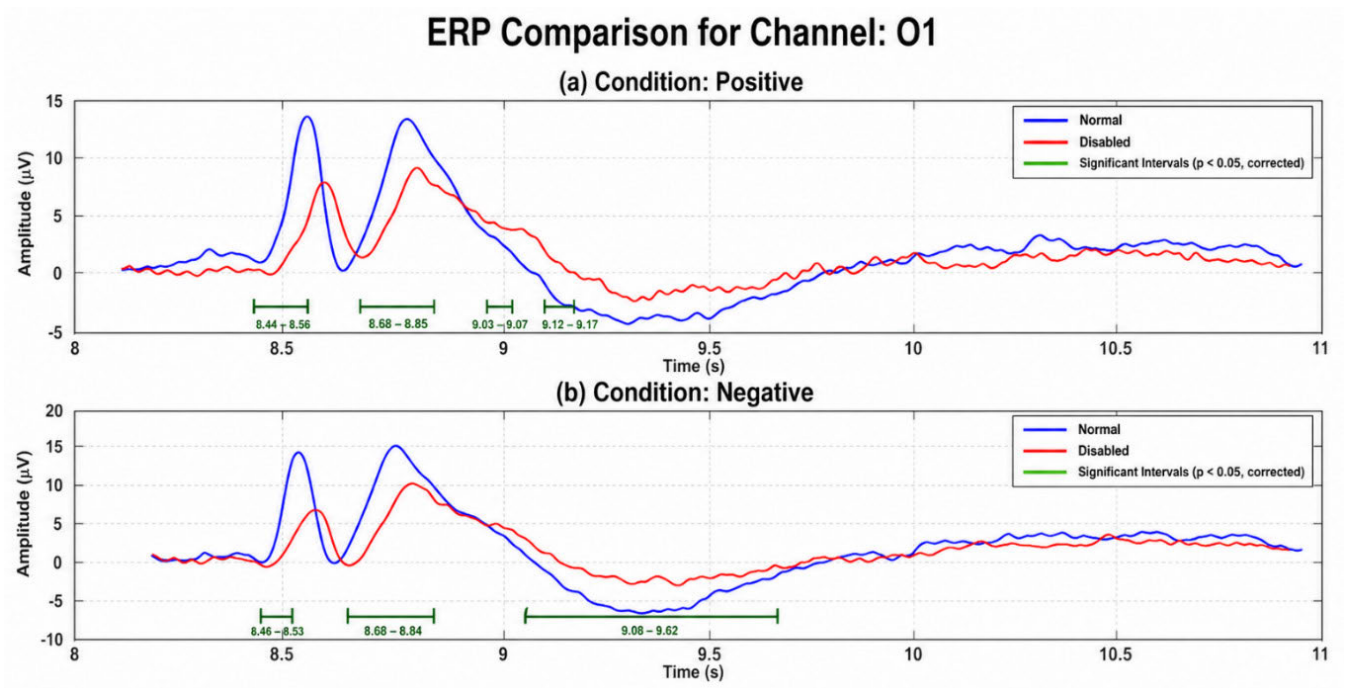


FIGURE 5. Comparison of ERP signals between the control and disorder groups at channel O1. The top panel shows the positive condition, and the bottom shows the negative. Green horizontal markers indicate time intervals with statistically significant differences between the groups.

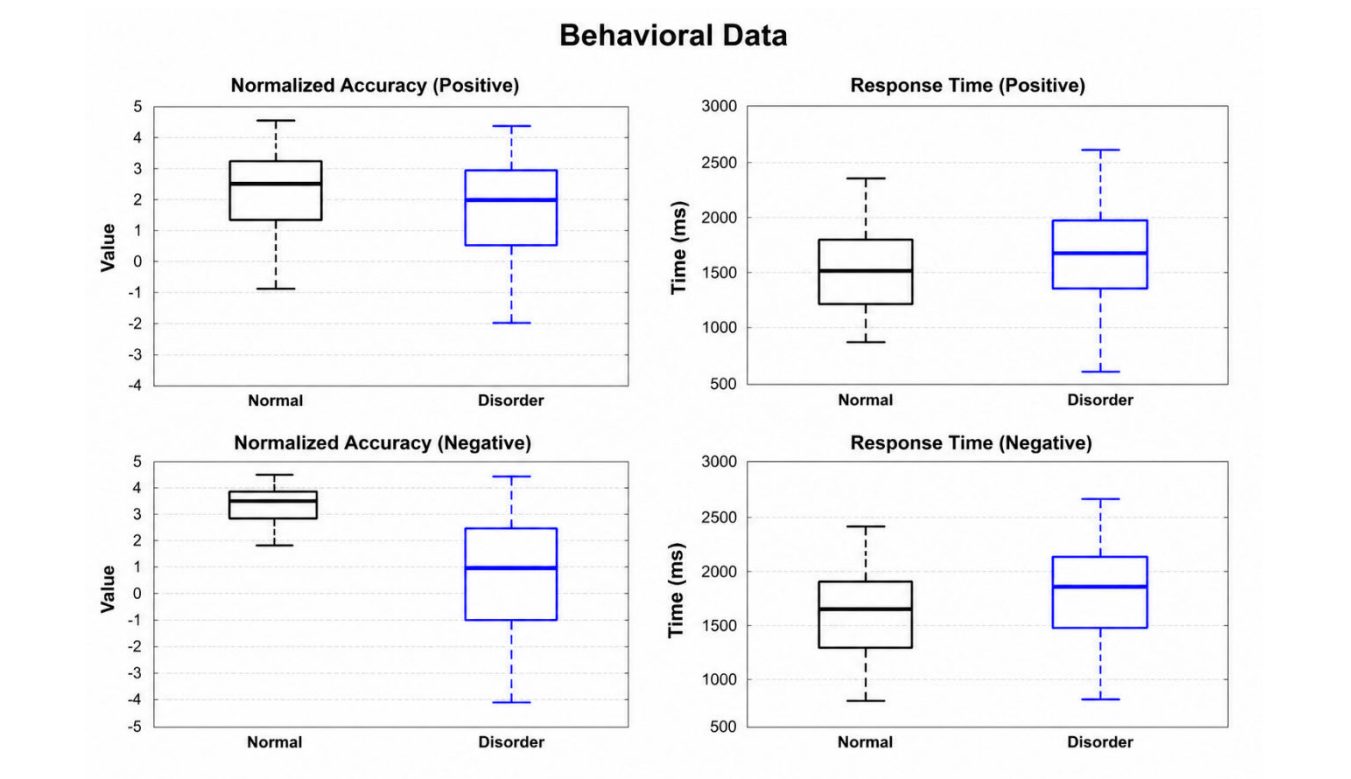


FIGURE 6. Boxplots of behavioral data. The top two plots correspond to the positive condition, and the bottom two to the negative condition.

enhanced efficacy in recognizing troubled individuals. The findings are displayed in Table 3.

By adding behavioral features to this expanded feature set, the results were mixed. While some combinations led

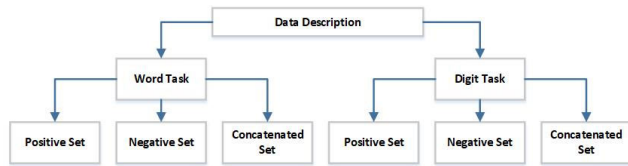


FIGURE 7. Different data for machine learning classification categories.

to minor accuracy gains, others caused performance to deteriorate. Table 4 shows the results.

(3) Full band features: To expand beyond the examination of individual components, additional features were extracted from the entire ERP. As follows, these features and other combinations are employed to classify: (1) One ERP component and EEG features; (2) One ERP component and EEG features and behavioral data; (3) Two ERP components and EEG features; and (4) Two ERP components and EEG features and behavioral data. The findings are illustrated in Table 5. The best performance ($86.36\% \pm 4.98\%$) was obtained for the numerical task (negative condition) using the third configuration. These results confirm that multi-component ERP features combined with global signal characteristics yield the most discriminative feature set. Behavioral features again showed inconsistent contributions, often failing to improve classifier performance.

(4) Feature selection methods: The entire feature set was utilized to evaluate six distinct methods of feature selection. This procedure was implemented to mitigate the risk of overfitting and address the high dimensionality of the extracted feature space. These were as follows: (1) no feature selection, (2) the univariate T-test, (3) the GA, (4) the PCA, (5) a combination of GA and PCA, and (6) a cascaded technique that

employed the T-test, GA, and PCA sequentially. To ensure that the generalization was valid and that information was not leaked, all methods were exclusively employed on the training data in each phase. On their own, the T-test and PCA consistently yielded the most accurate outcomes across all datasets. In the numerical task, the T-test method with FDR correction and a significance threshold of $p < 0.05$ correctly classified up to 81.53% of the cases when all conditions were combined. PCA performed similarly, attaining a maximal accuracy of 81.82%. In contrast, the GA was capable of identifying nonlinear relationships between features, despite its substantially inferior average performance. The best case achieved an accuracy of 78.64%. Occasionally, the combined methods, such as GA + PCA or T-test $\rightarrow$ GA $\rightarrow$ PCA, did not outperform the simpler methods. In certain instances, the additional complexity resulted in a decrease in accuracy, either due to overfitting or the loss of beneficial variance through unnecessary transformation processes. The detailed results are presented in Table 6.

V. DISCUSSION

This study presents a comprehensive EEG-based system for classifying children with LD. Non-invasive EEG signals were recorded during WM tasks, which include non-numerical and numerical tasks. A wide range of features was extracted and further integrated with behavioral data. Then, multiple feature selection strategies, including univariate filtering, evolutionary optimization, and dimensionality reduction techniques, were employed. Classification was performed using an SVM pipeline, incorporating Bayesian hyperparameter tuning and balanced cross-validation to ensure robustness.

Recent studies have explored EEG-based classification of learning disorders using a range of machine learning and deep

TABLE 1. Classification result employing one ERP component's features.

	NumericalTask			Non-numerical Task		
	Negative	Positive	Concatenate	Negative	Positive	Concatenate
Accuracy (%)	63.64 $\pm$ 4.98	60.91 $\pm$ 10.98	64.55$\pm$3.40	62.73 $\pm$ 4.45	63.64 $\pm$ 7.61	70.00$\pm$7.93
Sensitivity (0-1)	0.15 $\pm$ 0.15	0.30 $\pm$ 0.17	0.38 $\pm$ 0.34	0.25 $\pm$ 0.22	0.35 $\pm$ 0.22	0.40 $\pm$ 0.20
Specificity (0-1)	0.91 $\pm$ 0.08	0.79 $\pm$ 0.19	0.80 $\pm$ 0.23	0.84 $\pm$ 0.13	0.80 $\pm$ 0.12	0.87 $\pm$ 0.07
AUC (0-1)	0.68 $\pm$ 0.11	0.68 $\pm$ 0.07	0.62 $\pm$ 0.07	0.61 $\pm$ 0.11	0.63 $\pm$ 0.04	0.78 $\pm$ 0.05

TABLE 2. Classification result employing one ERP component's features and behavioral ones.

	NumericalTask			Non-numerical Task		
	Negative	Positive	Concatenate	Negative	Positive	Concatenate
Accuracy (%)	67.27 $\pm$ 6.03	77.27$\pm$5.75	66.36 $\pm$ 17.39	64.55$\pm$1.82	64.55 $\pm$ 6.68	60.91 $\pm$ 3.64
Sensitivity (0-1)	0.23 $\pm$ 0.17	0.50 $\pm$ 0.11	0.35 $\pm$ 0.25	0.15 $\pm$ 0.18	0.47 $\pm$ 0.15	0.30 $\pm$ 0.22
Specificity (0-1)	0.93 $\pm$ 0.08	0.93 $\pm$ 0.11	0.84 $\pm$ 0.19	0.93 $\pm$ 0.09	0.74 $\pm$ 0.16	0.79 $\pm$ 0.12
AUC (0-1)	0.68 $\pm$ 0.09	0.73 $\pm$ 0.11	0.65 $\pm$ 0.20	0.58 $\pm$ 0.06	0.67 $\pm$ 0.07	0.64 $\pm$ 0.07

TABLE 3. Classification result employing two ERPs components' features.

	NumericalTask			Non-numerical Task		
	Negative	Positive	Concatenate	Negative	Positive	Concatenate
Accuracy (%)	80.91±7.82	70.91±10.20	70.91±8.43	62.73±7.27	64.55±6.68	66.36±7.93
Sensitivity (0-1)	0.70±0.29	0.55±0.19	0.57±0.17	0.12±0.19	0.25±0.08	0.35±0.20
Specificity (0-1)	0.87±0.08	0.80±0.08	0.79±0.08	0.91±0.11	0.87±0.13	0.84±0.14
AUC (0-1)	0.82±0.10	0.76±0.10	0.79±0.07	0.56±0.12	0.77±0.11	0.70±0.07

TABLE 4. Classification result employing two ERPs components' features and behavioral data.

	NumericalTask			Non-numerical Task		
	Negative	Positive	Concatenate	Negative	Positive	Concatenate
Accuracy (%)	70.91±4.64	66.36±6.17	70.91±2.23	60.00±5.30	64.55±8.81	76.36±5.30
Sensitivity (0-1)	0.60±0.23	0.42±0.13	0.50±0.18	0.07±0.10	0.47±0.18	0.55±0.23
Specificity (0-1)	0.77±0.09	0.80±0.03	0.83±0.10	0.90±0.14	0.74±0.03	0.89±0.07
AUC (0-1)	0.75±0.09	0.72±0.03	0.84±0.04	0.57±0.09	0.68±0.10	0.79±0.04

learning methods. For example, Eroğlu et al. [41] proposed a machine learning framework using EEG frequency-band features for dyslexia prediction, demonstrating the effectiveness of spectral biomarkers. Similarly, a recent study [42] employed classical machine learning and deep learning

techniques on EEG-derived features for learning disability classification, achieving moderate classification performance. Another approach combined wavelet-based feature extraction with probabilistic classifiers to capture the time-frequency characteristics of EEG signals for dyslexia

TABLE 5. Classification result employing different feature vector combinations.

		NumericalTask			Non-numerical Task		
		Negative	Positive	Concatenate	Negative	Positive	Concatenate
1	Accuracy (%)	76.36±3.40	76.36±6.68	81.82±5.75	70.00±10.20	80.00±2.23	70.91±6.17
	Sensitivity (0-1)	0.68±0.13	0.65±0.18	0.65±0.09	0.40±0.22	0.68±0.10	0.45±0.06
	Specificity (0-1)	0.81±0.10	0.83±0.10	0.91±0.08	0.87±0.05	0.87±0.03	0.86±0.06
	AUC (0-1)	0.80±0.06	0.81±0.08	0.85±0.09	0.70±0.14	0.82±0.07	0.74±0.07
2	Accuracy (%)	80.00±2.23	72.73±4.98	78.18±7.82	67.27±6.03	80.91±8.81	77.27±4.98
	Sensitivity (0-1)	0.65±0.12	0.57±0.10	0.70±0.15	0.23±0.20	0.65±0.05	0.55±0.19
	Specificity (0-1)	0.89±0.07	0.81±0.07	0.83±0.11	0.93±0.05	0.90±0.13	0.90±0.06
	AUC (0-1)	0.83±0.04	0.76±0.11	0.85±0.07	0.81±0.11	0.84±0.08	0.81±0.06
3	Accuracy (%)	86.36±4.98	70.91±5.45	77.27±6.43	76.36±9.71	75.45±7.39	73.64±8.81
	Sensitivity (0-1)	0.72±0.12	0.65±0.09	0.70±0.17	0.72±0.17	0.62±0.19	0.62±0.22
	Specificity (0-1)	0.94±0.03	0.74±0.12	0.81±0.03	0.79±0.12	0.83±0.07	0.80±0.05
	AUC (0-1)	0.89±0.04	0.82±0.05	0.84±0.05	0.80±0.13	0.80±0.07	0.78±0.11
4	Accuracy (%)	71.82±6.03	70.00±7.39	86.36±4.07	63.64±6.43	74.55±5.82	72.73±6.74
	Sensitivity (0-1)	0.55±0.13	0.57±0.17	0.72±0.18	0.29±0.24	0.51±0.14	0.51±0.16
	Specificity (0-1)	0.81±0.07	0.77±0.03	0.94±0.05	0.84±0.15	0.88±0.07	0.85±0.07
	AUC (0-1)	0.80±0.07	0.74±0.13	0.90±0.06	0.63±0.12	0.83±0.07	0.79±0.07

TABLE 6. Classification result employing different feature selection methods.

		NumericalTask			Non-numerical Task		
		Negative	Positive	Concatenate	Negative	Positive	Concatenate
1 <i>None</i>	Accuracy (%)	74.98±2.27	74.84±2.08	77.02±3.28	63.16±2.58	63.41±1.4	74.25±3.02
	Sensitivity (0-1)	0.57±0.06	0.58±0.07	0.62±0.06	0.23±0.10	0.22±0.2	0.46±0.10
	Specificity (0-1)	0.85±0.02	0.85±0.02	0.86±0.03	0.87±0.04	0.88±0.8	0.91±0.02
	AUC (0-1)	0.82±0.03	0.80±0.02	0.84±0.03	0.65±0.06	0.62±0.62	0.81±0.04
2 <i>T-Test</i>	Accuracy (%)	74.09±2.29	73.29±4.32	81.53±3.05	72.8±3.98	74.85±3.71	80.31±2.99
	Sensitivity (0-1)	0.55±0.04	0.53±0.12	0.70±0.03	0.52±0.13	0.55±0.11	0.66±0.05
	Specificity (0-1)	0.85±0.02	0.85±0.03	0.88±0.04	0.85±0.05	0.87±0.04	0.89±0.04
	AUC (0-1)	0.81±0.02	0.80±0.06	0.90±0.03	0.77±0.06	0.80±0.05	0.89±0.02
3 <i>GA</i>	Accuracy (%)	73.64±5.68	76.36±9.27	78.64±6.44	68.64±6.25	73.64±8.33	72.73±6.43
	Sensitivity (0-1)	0.51±0.21	0.64±0.22	0.74±0.10	0.23±0.23	0.47±0.18	0.50±0.16
	Specificity (0-1)	0.86±0.08	0.84±0.11	0.81±0.13	0.95±0.07	0.89±0.11	0.86±0.11
	AUC (0-1)	0.78±0.08	0.82±0.09	0.84±0.07	0.76±0.08	0.82±0.09	0.79±0.08
4 <i>PCA</i>	Accuracy (%)	80.00±6.49	77.73±6.88	81.82±8.86	67.27±8.33	79.55±7.94	77.73±8.97
	Sensitivity (0-1)	0.75±0.12	0.62±0.16	0.66±0.17	0.26±0.26	0.66±0.16	0.49±0.15
	Specificity (0-1)	0.83±0.10	0.86±0.09	0.91±0.09	0.91±0.09	0.87±0.10	0.94±0.07
	AUC (0-1)	0.87±0.08	0.83±0.11	0.88±0.08	0.64±0.11	0.83±0.11	0.81±0.14
5 <i>GA+PCA</i>	Accuracy (%)	75.00±4.66	75.45±8.43	78.64±4.09	65.00±9.33	78.64±6.44	75.00±6.82
	Sensitivity (0-1)	0.55±0.13	0.55±0.15	0.68±0.11	0.23±0.24	0.60±0.22	0.39±0.21
	Specificity (0-1)	0.86±0.09	0.87±0.09	0.85±0.07	0.89±0.13	0.89±0.09	0.96±0.06
	AUC (0-1)	0.84±0.05	0.83±0.08	0.82±0.05	0.62±0.11	0.86±0.06	0.75±0.11
6 <i>T-Test+GA+PCA</i>	Accuracy (%)	70.45±7.11	78.64±9.55	74.55±6.17	71.82±6.98	70.45±6.51	76.36±6.98
	Sensitivity (0-1)	0.47±0.16	0.65±0.21	0.57±0.14	0.64±0.13	0.50±0.15	0.62±0.16
	Specificity (0-1)	0.84±0.09	0.86±0.07	0.84±0.09	0.76±0.09	0.82±0.08	0.84±0.12
	AUC (0-1)	0.75±0.07	0.82±0.10	0.86±0.06	0.77±0.09	0.75±0.07	0.83±0.05

detection [43]. These studies underscore the importance of feature engineering and signal representation but are generally limited to single-modality EEG features.

More recent work has also explored multimodal and large-scale frameworks. Sait et al. [44] introduced a transformer-based model integrating EEG, MRI, and handwriting data, achieving very high classification performance in multimodal settings. Additionally, Taskov et al. [45] investigated oscillatory EEG signatures associated with dyslexia, providing strong neurophysiological insights but without focusing on classification performance. In contrast to these studies, the proposed framework integrates ERP, spectral EEG, and behavioral features within a subject-independent evaluation pipeline applied to a relatively large cohort, providing a more realistic assessment of generalization performance while maintaining a balance between interpretability and practical applicability.

The results show that children with learning disabilities exhibited different cognitive responses in the later stages,

which may be linked to difficulties with attentional control. This is in line with findings from Cárdenas et al. [46], who reported diminished mid-latency ERP components in children diagnosed with dyscalculia. A study [47] found disrupted LPC responses during arithmetic conflict in LD children. The numerically based negative condition in this study, which elicited strong late ERPs, supports the notion that tasks involving cognitive load and conflict yield more distinct neural signatures. Moreover, integrating ERP features with EEG features, such as frequency-band power, entropy, ERS/ERD, and theta–gamma PAC, significantly improved classification performance. This finding underscores the advantages of integrating complementary neural representations. The use of PAC in this study is consistent with findings in [48], which employed PAC features for dyslexia classification. The current framework involved the analysis of PAC in conjunction with ERS/ERD, entropy, and spectral power features. This integration enhances the robustness of the multimodal classification model and aligns with

broader research in mathematical cognition; for instance, oscillatory EEG studies indicate that theta and alpha desynchronization correlate with heightened arithmetic demands in children, as noted by [49]. Torres-Ramos et al. [50] employed graph metrics based on coherence across frequency bands to classify the mathematical achievement of primary school children using EEG, attaining accuracies of up to 80%. The numerical-based paradigm demonstrated the highest classification accuracy under negative conditions, achieving up to 86.36% when integrating two ERP components with EEG features. This finding indicates that numerical stimuli, which likely impose a higher cognitive load on children with dyscalculia, generate more distinguishable neural patterns than non-numerical-based tasks. Although behavioral features such as reaction time and accuracy showed clear group-level differences, their inclusion did not consistently improve classification outcomes. The comparative analysis of different feature configurations indicates that the integration of ERP components with broader EEG-derived features yields the most effective representation for classification. This is reflected in the highest achieved performance. Consistently, behavioral features, although showing clear group-level differences, contributed only marginally when combined with neural features. Furthermore, the results demonstrate that simpler feature selection approaches such as T-test and PCA can achieve competitive performance, suggesting that effective classification can be obtained without relying on highly complex or high-dimensional feature sets. In terms of feature selection, simpler strategies such as univariate T-tests and PCA outperformed more complex methods (e.g., GA + PCA), offering a better balance between performance and interpretability. It should be noted that the inclusion of multiple feature selection strategies was intended to enable a systematic comparison within a unified framework. The strong performance of simpler methods highlights their effectiveness and interpretability in this setting, while the inclusion of more complex approaches such as GA provides additional flexibility for exploring alternative configurations in future studies. The GA, despite its theoretical strengths in modeling nonlinear dependencies, introduced variability and was less stable across repetitions. A consistent pattern across all methods was that specificity was higher than sensitivity, indicating that the models were more effective at identifying typically developing children than detecting those with LD. This pattern is consistent with the inherent challenges of EEG-based classification in neurodevelopmental disorders, where neural signatures often overlap between groups, particularly in heterogeneous or mild cases. In addition, the dataset exhibits moderate class imbalance, which may bias the classifier toward the majority class and contribute to higher specificity. From a practical perspective, techniques such as decision threshold adjustment or cost-sensitive learning could be applied to improve sensitivity, particularly in clinical applications where minimizing false negatives is critical. All classification results are reported as mean $\pm$ standard deviation over 10 independent iterations to reflect performance

stability. Although this repeated evaluation provides a robust estimate of model behavior, formal statistical hypothesis testing between different feature combinations was not conducted in this study and remains an important direction for future work. Future work will include formal statistical hypothesis testing (e.g., paired statistical comparisons across repeated runs) to quantitatively assess differences between feature configurations and further strengthen the robustness of the conclusions.

Although the evaluation was conducted on a single cohort, which limits direct assessment of cross-dataset generalization, a subject-independent and leakage-free framework was employed to provide a more realistic estimate of generalization performance. By separating training and testing data at the subject level, the model avoids learning subject-specific patterns and instead captures more generalizable neural features. While these results suggest that the proposed framework has the potential to generalize to new subjects, further validation on independent datasets and across different recording conditions will be necessary to confirm its robustness in broader clinical settings.

The present findings support the feasibility of a comprehensive EEG-based framework for the classification of developmental learning disorders, combining a working memory task with multimodal neural and behavioral analysis. By incorporating both non-numerical and numerical stimuli, the task facilitated the extraction of ERP features and behavioral measures within a subject-independent, automated pipeline. The approach achieved promising accuracy and interpretability, particularly when late ERP components and spectral dynamics were included. However, the relatively lower sensitivity highlights the need for further improvements in task design and feature representation to better capture subtle neural signatures of learning difficulties. Advancing both cognitive paradigms and classification strategies will be essential to develop clinically reliable, scalable diagnostic tools for learning disorders across diverse real-world settings.

VI. CONCLUSION

The presented study introduced a comprehensive EEG-based framework for classifying children with LD. It combined ERP components, EEG features, and behavioral performance metrics to create an integrated diagnostic pipeline. The results show that late-stage ERP components are effective. The numerical task under negative stimulus conditions consistently yielded the highest classification accuracy. Behavioral measures showed differences between groups but contributed minimally to classification accuracy when paired with EEG features. Despite the study's strengths, two methodological constraints should be acknowledged. First, although ERP time windows were defined based on prior statistical analysis of the same dataset, window selection was conducted independently of the machine-learning procedure, preventing leakage. Second, while repeated subject-independent splits provide robust internal estimates, external validation on an independent cohort remains an important direction for

future work. For future studies, designing more cognitively demanding, task-specific paradigms is suggested. Integrating advanced dynamic features, such as graph-theoretic connectivity, can uncover more subtle patterns in EEG.

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REFERENCES

- [1] A. J. Wilson, S. G. Andrewes, H. Struthers, V. M. Rowe, R. Bogdanovic, and K. E. Waldie, "Dyscalculia and dyslexia in adults: Cognitive bases of comorbidity," *Learn. Individual Differences*, vol. 37, pp. 118–132, Jan. 2015.
- [2] W. Han, "Dyscalculia and dyslexia in school-aged children: Comorbidity, support, and future prospects," *Frontiers Educ.*, vol. 10, Jan. 2025, Art. no. 1515216.
- [3] Z. Cui, Z. Xia, M. Su, H. Shu, and G. Gong, "Disrupted white matter connectivity underlying developmental dyslexia: A machine learning approach," *Human Brain Mapping*, vol. 37, no. 4, pp. 1443–1458, Apr. 2016.
- [4] M. Rosenberg-Lee, S. Ashkenazi, T. Chen, C. B. Young, D. C. Geary, and V. Menon, "Brain hyper-connectivity and operation-specific deficits during arithmetic problem solving in children with developmental dyscalculia," *Develop. Sci.*, vol. 18, no. 3, pp. 351–372, May 2015.
- [5] R. M. Rangayyan and S. Krishnan, *Biomedical Signal Analysis*. Hoboken, NJ, USA: Wiley, 2024.
- [6] C. Turri, G. Di Dona, A. Santoni, D. A. Zamfira, L. Franchin, D. Melcher, and L. Ronconi, "Periodic and aperiodic EEG features as potential markers of developmental dyslexia," *Biomedicine*, vol. 11, no. 6, p. 1607, Jun. 2023.
- [7] M. A. Formoso, A. Ortiz, F. J. Martinez-Murcia, N. Gallego, and J. L. Luque, "Detecting phase-synchrony connectivity anomalies in EEG signals. Application to dyslexia diagnosis," *Sensors*, vol. 21, no. 21, p. 7061, Oct. 2021.
- [8] A. Mingozzi, V. Tobia, and G. M. Marzocchi, "Dyslexia and dyscalculia: Which neuropsychological processes distinguish the two developmental disorders?" *Child Neuropsychology*, vol. 30, no. 1, pp. 1–21, Jan. 2024.
- [9] A. Premeti, M. P. Bucci, and F. Isel, "Evidence from ERP and eye movements as markers of language dysfunction in dyslexia," *Brain Sci.*, vol. 12, no. 1, p. 73, Jan. 2022.
- [10] A. Premeti, M. P. Bucci, K. Heidlmayr, P. Vigneron, and F. Isel, "Neurodynamics of selected language processes involved in word reading: An EEG study with French dyslexic adults," *J. Neurolinguistics*, vol. 71, Aug. 2024, Art. no. 101201.
- [11] N. P. G. Seshadri, B. K. Singh, and R. B. Pachori, "EEG based functional brain network analysis and classification of dyslexic children during sustained attention task," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 31, pp. 4672–4682, 2023.
- [12] D. Devi and S. Sophia, "GA-CNN: Analyzing student's cognitive skills with EEG data using a hybrid deep learning approach," *Biomed. Signal Process. Control*, vol. 90, Apr. 2024, Art. no. 105888.
- [13] S. Shalileh, D. Ignatov, A. Lopukhina, and O. Dragoy, "Identifying dyslexia in school pupils from eye movement and demographic data using artificial intelligence," *PLoS ONE*, vol. 18, no. 11, Nov. 2023, Art. no. e0292047.
- [14] N. P. G. Seshadri, B. Geethanjali, and B. K. Singh, "EEG based functional brain networks analysis in dyslexic children during arithmetic task," *Cognit. Neurodynamics*, vol. 16, no. 5, pp. 1013–1028, Oct. 2022.
- [15] N. P. G. Seshadri, S. Agrawal, B. K. Singh, B. Geethanjali, V. Mahesh, and R. B. Pachori, "EEG based classification of children with learning disabilities using shallow and deep neural network," *Biomed. Signal Process. Control*, vol. 82, Apr. 2023, Art. no. 104553.
- [16] A. Oliaee, M. Mohebbi, S. Shirani, and R. Rostami, "Extraction of discriminative features from EEG signals of dyslexic children: before and after the treatment," *Cognit. Neurodynamics*, vol. 16, no. 6, pp. 1249–1259, Dec. 2022.
- [17] A. K. Dewangan, N. P. G. Seshadri, B. K. Singh, G. Balasubramaniam, and M. Veezhinathan, "Determining dominant EEG channels for classification of LD and non-LD children using machine learning approach," in *Proc. 13th Int. Conf. Softw. Comput. Appl.*, Feb. 2024, pp. 1–6.
- [18] J. Jincy and P. S. H. Jose, "Preliminary study of dyslexia using machine learning algorithms," *AIP Conf. Proc.*, vol. 3122, 2024, Art. no. 030005.
- [19] M. Chalmpé and F. Vlachos, "Working and long-term memory deficits in developmental dyslexia," in *Proc. INTED*, vol. 1, 2025, pp. 3482–3487.
- [20] M. Nakamura and E. Rossi, "Differential effects of working memory constraints on syntactic and semantic processing: Evidence from ERPs," *Brain Lang.*, vol. 264, May 2025, Art. no. 105550.
- [21] B. J. Martínez-Briones, T. Fernández, and J. Silva-Pereyra, "Electroencephalographic power spectrum patterns related to the intelligence of children with learning disorders," *PeerJ*, vol. 13, Mar. 2025, Art. no. e19138.
- [22] V. Manimaran, N. Shanthi, N. Aravindhraj, K. Rakesh, K. Revanth, and S. Saran, "Enhancing dyslexia detection: A robust machine learning approach," in *Proc. 3rd Int. Conf. Intell. Data Commun. Technol. Internet Things (IDCIOT)*, Feb. 2025, pp. 1517–1523.
- [23] E. Cainelli, L. Vedovelli, B. Carretti, and P. Bisiacchi, "EEG correlates of developmental dyslexia: A systematic review," *Ann. Dyslexia*, vol. 73, no. 2, pp. 184–213, 2023. [Online]. Available: <https://doi.org/10.1007/s11881-022-00273-1>
- [24] I. Mikheev, H. Steiner, and O. Martynova, "Detecting cognitive traits and occupational proficiency using EEG and statistical inference," *Sci. Rep.*, vol. 14, no. 1, p. 5605, Mar. 2024.
- [25] G. Guerra, A. Tierney, J. Tijms, A. Vaessen, M. Bonte, and F. Dick, "Attentional modulation of neural sound tracking in children with and without dyslexia," *Develop. Sci.*, vol. 27, no. 1, p. 13420, Jan. 2024.
- [26] S. Zhong, S. Song, T. Tang, F. Nie, X. Zhou, Y. Zhao, Y. Zhao, K. F. Sin, and S.-H.-G. Chan, "DYPA: A machine learning dyslexia prescreening mobile application for Chinese children," *Proc. ACM Interact., Mobile, Wearable Ubiquitous Technol.*, vol. 7, no. 3, pp. 1–21, Sep. 2023.
- [27] R. Vaitheeshwari, C.-H. Chen, C.-R. Chung, H.-Y. Yang, S.-C. Yeh, E. H.-K. Wu, and M. Kumar, "Dyslexia analysis and diagnosis based on eye movement," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 32, pp. 4109–4119, 2024.
- [28] J. Sprafkin, K. D. Gadow, H. Salisbury, J. Schneider, and J. Loney, "Further evidence of reliability and validity of the child symptom inventory-4: Parent checklist in clinically referred boys," *J. Clin. Child Adolescent Psychol.*, vol. 31, no. 4, pp. 513–524, Nov. 2002.
- [29] E. Mohammedsmaeil and H. A. J. O. E. C. Hooman, "Adaptation and standardization of the IRAN KEY-MATH test of mathematics," vol. 2, no. 4, pp. 323–332, 2003.
- [30] E. G. Willcutt, R. Boada, M. W. Riddle, N. Chhabildas, J. C. DeFries, and B. F. Pennington, "Colorado learning difficulties questionnaire: Validation of a parent-report screening measure," *Psychol. Assessment*, vol. 23, no. 3, pp. 778–791, 2011.
- [31] D. Wechsler, "Wechsler intelligence scale for children," WISC-III, 3rd ed., Manual. San Antonio, TX, USA: The Psychological Corporation, 1991.
- [32] M. A. Nazari et al., "Iranian 6–11 years age population-based EEG, ERP, and cognition dataset," *Sci. Data*, vol. 12, no. 1, p. 319, Feb. 2025.
- [33] A. Delorme and S. Makeig, "EEGLAB: An open source toolbox for analysis of single-trial EEG dynamics including independent component analysis," *J. Neurosci. Methods*, vol. 134, no. 1, pp. 9–21, Mar. 2004.
- [34] M. Plechawska-Wójcik, M. Kaczorowska, and D. Zapala, "The artifact subspace reconstruction (ASR) for EEG signal correction. A comparative study," in *Proc. Int. Conf. Inf. Syst. Archit. Technol.*, 2018, pp. 125–135.
- [35] L. Pion-Tonachini, K. Kreutz-Delgado, and S. Makeig, "ICLabel: An automated electroencephalographic independent component classifier, dataset, and website," *NeuroImage*, vol. 198, pp. 181–197, Sep. 2019.
- [36] G. Pfurtscheller and F. H. Lopes da Silva, "Event-related EEG/MEG synchronization and desynchronization: Basic principles," *Clin. Neurophysiol.*, vol. 110, no. 11, pp. 1842–1857, Nov. 1999.
- [37] R. T. Canolty, E. Edwards, S. S. Dalal, M. Soltani, S. S. Nagarajan, H. E. Kirsch, M. S. Berger, N. M. Barbaro, and R. T. Knight, "High gamma power is phase-locked to theta oscillations in human neocortex," *Science*, vol. 313, no. 5793, pp. 1626–1628, Sep. 2006.
- [38] H. G. Schaathun, "Support vector machines," in *Machine Learning Models and Algorithms for Big Data Classification: Thinking With Examples for Effective Learning*. Cham, Switzerland: Springer, 2012, pp. 179–196.
- [39] A. Lambora, K. Gupta, and K. Chopra, "Genetic algorithm—A literature review," in *Proc. Int. Conf. Mach. Learn., Big Data, Cloud Parallel Comput. (COMITCon)*, Feb. 2019, pp. 380–384.

- [40] H. Abdi and L. J. J. W. I. R. C. S. Williams, "Principal component analysis," *WIREs Comput. Statist.*, vol. 2, no. 4, pp. 433–459, 2010.
- [41] G. Eroğlu and M. R. Abou Harb, "Theta and Beta1 frequency band values predict dyslexia classification," *Dyslexia*, vol. 32, no. 1, p. 70021, Feb. 2026.
- [42] S. Khalfallah, W. Puech, M. Tlija, and K. Bouallegue, "Exploring the effectiveness of machine learning and deep learning techniques for EEG signal classification in neurological disorders," *IEEE Access*, vol. 13, pp. 17002–17015, 2025.
- [43] A. K. Dewangan, B. K. Singh, and G. Sheshadri, "DWT based classification of children with dyslexia from normal by using SVM classifier on EEG signals," in *Proc. Int. Conf. Biomed. Eng. Sci. Technol.*, 2026, pp. 634–653.
- [44] A. R. W. Sait and Y. Alkhurayyif, "Lightweight hybrid transformers-based dyslexia detection using cross-modality data," *Sci. Rep.*, vol. 15, no. 1, p. 17054, May 2025.
- [45] T. Taskov and J. Dushanova, "Functional networks in developmental dyslexia: Auditory discrimination of words and pseudowords," *NeuroSci*, vol. 7, no. 1, p. 21, Feb. 2026.
- [46] S. Y. Cárdenas, J. Silva-Pereyra, B. Prieto-Corona, S. A. Castro-Chavira, and T. Fernández, "Arithmetic processing in children with dyscalculia: An event-related potential study," *PeerJ*, vol. 9, Jan. 2021, Art. no. e10489.
- [47] J. Araújo, B. D. Simons, V. Peter, K. Mandke, M. Kalashnikova, A. Macfarlane, F. Gabrielczyk, A. Wilson, G. M. Di Liberto, D. Burnham, and U. Goswami, "Atypical low-frequency cortical encoding of speech identifies children with developmental dyslexia," *Frontiers Human Neurosci.*, vol. 18, Jun. 2024, Art. no. 1403677.
- [48] N. J. Gallego-Molina, A. Ortiz, F. J. Martínez-Murcia, M. A. Formoso, and A. Giménez, "Complex network modeling of EEG band coupling in dyslexia: An exploratory analysis of auditory processing and diagnosis," *Knowl.-Based Syst.*, vol. 240, Mar. 2022, Art. no. 108098.
- [49] C. Artemenko, M. Soltanlou, S. M. Bieck, A.-C. Ehliis, T. Dresler, and H.-C. Nuerk, "Individual differences in math ability determine neurocognitive processing of arithmetic complexity: A combined fNIRS-EEG study," *Frontiers Human Neurosci.*, vol. 13, p. 227, Jul. 2019.
- [50] S. Torres-Ramos, R. A. Salido-Ruiz, A. Espinoza-Valdez, F. R. Gómez-Velázquez, A. A. González-Garrido, and I. Román-Godínez, "A brain connectivity characterization of children with different levels of mathematical achievement based on graph metrics," *PLoS ONE*, vol. 15, no. 1, Jan. 2020, Art. no. e0227613.



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